Sasan Maleki

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Summary

Strategic Data Science & AI Leader (PhD in Computer Science) with a strong focus on the fintech industry, adept at building and guiding high-performing teams to deliver advanced AI and data-driven solutions. Proven expertise in developing and deploying impactful AI-driven systems for BNPL credit scoring, KYC automation, and fraud detection, combining deep technical knowledge with entrepreneurial success.

Professional Experience

Tara
07/2022 - Present

Head of Data Science & AI

- Established the Data Science & AI team from the ground up, fostering a team culture focused on developing innovative AI products leveraging data from millions of users and hundreds of merchants.
- Directed the strategy, design, and development of AI-driven BNPL scoring models, substantially reducing the default rate and late payments.
- Led the development of automated Know Your Customer (KYC) solutions, reducing manual processing by up to 90% and improving customer onboarding efficiency.
- Spearheaded the development and implementation of fraud detection systems.
- Introduced and scaled critical data infrastructure, including data warehouses, BI dashboards, and automated alert systems, to facilitate company growth and operational efficiency.
- Championed data-driven decision-making across the organization by overseeing critical data analysis projects.

Perpetual Labs
07/2021 - 10/2021

Machine Learning Consultant
- Provided strategic consultancy and technical leadership in implementing advanced algorithms for automatic machining feature detection, aiming to enhance manufacturing process automation.

Pixalione
05/2021 - 08/2021

Machine Learning Consultant
- Advised on the strategic application of machine learning techniques to significantly improve search engine optimisation, increasing organic traffic for key client projects.

Freelance
06/2018 - 04/2021

Data Scientist & IoT Solutions Architect

- Architected and delivered end-to-end data science and IoT solutions for diverse clients, managing project lifecycles from initial design and hardware manufacturing to algorithm deployment and performance monitoring.
- Led client engagements to define project scope, translating complex business requirements into technical specifications for projects including industrial IoT predictive control systems, and analytics platforms].
- Developed and implemented deep learning anomaly detection algorithms for temperature sensor data in gas turbines.
- Engineered vehicle traffic prediction models.
- Designed and oversaw the manufacturing of smart electric heater controls, integrating hardware and software for optimized performance.

ECS Partners Ltd
Jan 2014 - Sep 2014

Researcher

- Developed optimisation algorithms for coordinating cooling loads of groups of apartments, and devised a time-efficient cost division scheme for dividing the shared cost of cooling in a fair way.

British Telecom (BT)
01/2011 - 06/2011

Researcher and Lead Developer

- Led the design and development of an innovative web-based platform for British Telecom (BT), a major UK telecommunications provider, to measure and enhance employee awareness of energy consumption at work and home.
- Engineered the system to gamify employee interaction, effectively populating a rich dataset for scientific research into energy usage patterns and validating hypotheses on behavioral change.
- Managed the project from initial research and concept through to prototype deployment, aligning technical development with research objectives.

Ventures

Gravity Tech
2020

Founder

- Founded Gravity Tech, a spin-off venture specializing in the design and manufacturing of custom electronic boards for industrial coffee roasting machines.
- Engineered electronics solutions to connect roasting machinery to computer systems for real-time monitoring and automated control of actuators based on sensor data.
- Pioneered technology aimed at significantly reducing operational costs and manual labor in the coffee roasting industry.

Behzisaz
06/2021 - Present

Co-founder

- Co-founded Behzisaz, a manufacturer of customized laboratory equipment and rapid diagnostic tests.

- Led the design, development, and manufacturing of an innovative smart medical pump capable of high-precision liquid handling (injection and withdrawal) at nanoliter scales.
- Engineered the device to automate complex and lengthy experiments in biology and medical labs by executing predefined programs, significantly enhancing experimental accuracy and efficiency.

Education

Sharif University of Technology
07/2017 - 07/2018

Postdoctoral Research Fellow
- Developed optimisation algorithms for coordinating cooling loads of groups of apartments
- Devised a fair and time efficient algorithm for dividing the shared cost of coordinated cooling

University of Southampton, UK
06/2011 - 09/2015

PhD Computer Science
Thesis: Addressing the Computational Issues of the Shapley Value With Applications in the Smart Grid
Supervisors: Prof. Alex Rogers, Dr. Talal Rahwan

University of Southampton, UK
09/2009 - 09/2010

MSc Artificial Intelligence (with Distinction)
Thesis: Gaussian Process Regression for Carbon Emissions from Heating and Cooling Loads
Supervisor: Prof. Alex Rogers

Heriot-Watt University, UK
IASBS, Iran
2004 - 2009

BSc Information Technology

Awards

National Elites Foundation
2017

Postdoctoral Fellowship
Iran

University of Southampton
2011

Fully Funded PhD Scholarship
United Kingdom

University of Southampton
2010

Distinction in Master's Degree
United Kingdom

INVITED TALKS

The 17th International Conference of Operations Research Society
July 2024

Optimal Credit Scoring

Sharif University of Technology
March 2019

Introduction to Kalman Filtering

Sharif University of Technology
October 2018

Forecasting Using Gaussian Process Regression

Sharif University of Technology
December 2016

An Intelligent Agent For Home Heating,
School of Electrical Engineering

Sharif University of Technology
November 2016

Cooperative Games and the Problem of Dividing the Payoff,
School of Mathematical Sciences

IASBS
May 2016

Artificial Intelligence Challenges in the Smart Grid

Zanjan University
May 2016

Artificial Intelligence Challenges in the Smart Grid